

JAMES RUSE AGRICULTURAL HIGH SCHOOL

YEAR 11 PRELIMINARY EXAM

1999

MATHEMATICS

The exam is divided into two sections; A and B

2 Unit students :

The 2 Unit students have 90 minutes to complete section A. After 90 minutes the 2 Unit students must leave the hall (not any earlier).

3 Unit students :

The 3 Unit students have 150 minutes to complete both sections A and B.

INSTRUCTIONS:

- 1.** Start each new question on a separate sheet of paper
- 2.** Show all necessary working out and formulae
- 3.** Approved calculators may be used
- 4.** Your student number must be on every page
- 5.** Marks may be deducted for untidy work
- 6.** Every question in both sections is worth 15 marks.
- 7.** The answers to the questions are to be returned in separate bundles.

SECTION A

QUESTION ONE

- a) Factorise $8 - 2x - 3x^2$
- b) Solve: i) $|x - 2| = 6$
- ii) $2^{2p} \cdot 3^p = 144$
- c) i) Express the following equation of a circle in the form $(x - a)^2 + (y - b)^2 = r^2$
 $x^2 + y^2 - 4x + 6y + 3 = 0$
- ii) Write down the coordinates of the circle's centre and the length of its radius.
- d) Simplify $(3 + \sqrt{2})^2 - (3 - \sqrt{2})^2$
- e) Find the area of the sector, which subtends an angle of 112° at the centre of a circle with radius 5cm.

QUESTION TWO

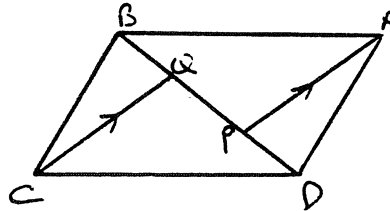
- a) Differentiate:
- i) $y = e^{6-2x}$
- ii) $y = x \cdot \tan x$
- iii) $y = \ln(5x + 1)$
- b) Write down the domain and range for the following functions:
- i) $y = 3^x$
- ii) $y = \sqrt{x+3}$
- c) A particle is traveling such that its displacement in metres at any time t seconds is given by $x = \sin\left(\frac{\pi}{2}t\right)$. Given that the particle starts from rest at the origin find:
- i) the direction the particle first moves in
- ii) when it next comes to rest
- iii) how far the particle travels in the first three seconds.

QUESTION THREE

- a) Find the equation of the tangent to the curve $y = x^2 - 2x - 4$ at the point where $x = 1$.
- b) For the parabola $y = x^2 - 2x - 4$, write down the:
- Coordinates of the vertex
 - Coordinates of the focus
 - equation of the directrix
- c) A surveyor travels from A, on a bearing of 130° , to a point B 8km from A. She then travels on a bearing of 030° to a point C, due East of A. Find the distance AC in kilometres correct to two decimal places.
- d) S_n denotes the sum of the first n terms of a Geometric series. If $s_n = \frac{a^n - b^n}{b^n - 1}$ find the first term and the common ratio of the series in terms of a and b .

QUESTION FOUR

- a) Find the set of values of k for which the roots of $2x^2 + kx + 3k - 10 = 0$ are real.
- b) ABCD is a parallelogram
AP//CQ
Prove that $\triangle ADP \equiv \triangle CBQ$.



- c) A variable cylinder of radius r units and height h units is inscribed in a fixed cone, radius a units and height b units.
- Prove that $\frac{b}{a} = \frac{h}{a-r}$
 - Express the volume of the cylinder as a function of r .
 - Prove that the maximum volume of the cylinder is $\frac{4}{9}$ of the cone's volume.

END OF SECTION A

SECTION B

QUESTION FIVE

- a). When $P(x) = x^3 + 2x^2 - kx + 5$ is divided by $(x + 3)$ the remainder is 1, find the value of k .
- b) Sketch the graph of $y = x(x - 1)^2$ without the use of calculus.
- c). Prove that $\frac{1}{\tan A + \cot B} + \frac{1}{\cot A + \tan B} = \frac{\sin(A + B)}{\cos(A - B)}$
- d) Find the exact value of $\tan 75^\circ$
- e) Find $\lim_{x \rightarrow -3} \frac{2x^3 + 54}{x + 3}$

QUESTION SIX

- a) If $f(x) = x^4 + bx^2 + cx + d$, find the values of b , c and d such that $f(x)$ is an even function, $f(0) = 1$ and $f(1) = 0$.
- b) A meeting room contains a round table surrounded by 10 chairs. These chairs are indistinguishable and equally spaced around the table. Find the number of different arrangements possible if:
- i) there are no restrictions
 - ii) 3 particular people must sit together
 - iii) Person A must not sit next to person B.
- c) Solve $2 \sin \theta - \cos \theta = 1$ for $0 \leq \theta \leq \pi$ using the t- results.
- d) The lengths of the sides of a scalene triangle are in an arithmetic sequence. If the largest angle in such a triangle is 120° , show that the smallest angle is $21^\circ 47'$.

1999 Preliminary Exam

SECTION A

Q1. (15)

a) $8 - 2x - 3x^2$

$$\begin{aligned} x^{-2} &= 8 - 6x + 4x - 3x^2 \\ &= 2(4 - 3x) + x(4 - 3x) \\ &= (2 + x)(4 - 3x) \end{aligned}$$

b) i) $|x - 2| = 6$

$$\begin{aligned} (x - 2 = 6 \quad \text{or} \quad -x + 2 = 6 \\ \underline{x = 8} \quad \text{or} \quad \underline{x = -4} \end{aligned}$$

ii) $2^{2p} \cdot 3^p = 144$

$$2^{2p} \cdot 3^p = 2^4 \cdot 3^2$$

$$\therefore p = 2$$

c) i) $x^2 - 4x + y^2 + 6y = -3$

$$(x - 2)^2 + (y + 3)^2 = -3 + 4 + 9$$

$$(x - 2)^2 + (y + 3)^2 = 10$$

ii) Centre is $(2, -3)$
radius is $\sqrt{10}$ units

d) $(3 + \sqrt{2} + 3 - \sqrt{2})(3 + \sqrt{2} - 3 + \sqrt{2})$

$$= 6 \times 2\sqrt{2}$$

$$= 12\sqrt{2}$$

e) $112^\circ = \frac{28\pi}{45}$

$$A = 25 \times \frac{28\pi}{45} = 48.9 \text{ cm}^2$$

Q2. (15)

a) i) $y = e^{6-2x}$

$$\frac{dy}{dx} = -2e^{6-2x}$$

ii) $y = x \cdot \tan x$

$$\frac{dy}{dx} = x \sec^2 x + \tan x$$

iii) $y = \ln(5x + 1)$

$$\frac{dy}{dx} = \frac{5}{5x + 1}$$

b) i) $y = 3^x$

Domain is all real x

Range is $y > 0$

ii) $y = \sqrt{x + 3}$

Domain is $x \geq -3$

Range is $y \geq 0$

c) i) $x = \sin\left(\frac{\pi}{2}t\right)$

$$v = \frac{\pi}{2} \cos\left(\frac{\pi}{2}t\right)$$

when $t = 0, v = \frac{\pi}{2}$

∴ first moves to the right

ii) at rest when $v = 0,$

$$\text{i.e. } \cos\left(\frac{\pi}{2}t\right) = 0$$

$$\frac{\pi}{2}t = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$t = 1, 3$$

∴ next comes to rest when $t = 1,$

iii) 3 units.

Q3 (13)
a) $y = x^2 - 2x - 4$
 $\frac{dy}{dx} = 2x - 2$

when $x=1, y=0$

$y = -5$

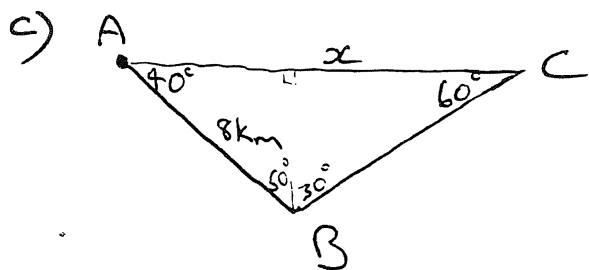
$\therefore$ eqn. of tangent is $y = -5$

b) i) $y + 4 = x^2 - 2x$
 $y + 5 = (x - 1)^2$

vertex is $(1, -5)$

ii) focus $(1, -4\frac{3}{4})$

iii) Directrix $y = -5\frac{1}{4}$



$$\frac{x}{\sin 80^\circ} = \frac{8}{\sin 60^\circ}$$

$$x = \frac{8 \times \sin 80^\circ}{\sin 60^\circ}$$

$$= 9.097264341$$

$$\approx 9.10 \text{ km}$$

d) $S_n = \frac{a^n - b^n}{b^n - 1}$

$S_1 = a - b$ $\therefore$ first term is $(a - b)$

$$S_2 = \frac{a^2 - b^2}{b}$$

$$S_2 - S_1 = T_2$$

$$\therefore T_2 = \frac{a^2 - b^2}{b} - a - b$$

$$= \frac{a^2 - ab}{b}$$

ratio is $\frac{a(a-b)}{b} \times \frac{1}{a-b} = \frac{a}{b}$

Q1 (15)

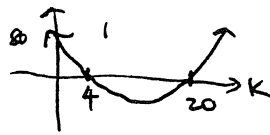
a) $2x^2 + Kx + 3K - 10 = 0$

$$b^2 - 4ac \geq 0$$

$$K^2 - 8(3K - 10) \geq 0$$

$$K^2 - 24K + 80 \geq 0$$

$$(K - 20)(K - 4) \geq 0$$



$$K \leq 4, K \geq 20$$

b)

$\angle APQ = \angle CQP$ (alternate angles; $CB \parallel DA$)
 $\therefore 180^\circ - \angle APQ = 180^\circ - \angle CQP$
 $\therefore \angle APD = \angle BQC$ (adjacent angles on a straight line)

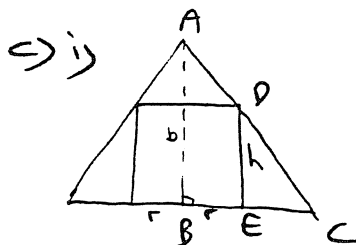
$\therefore$ In Δ 's ADP, CBQ

$\therefore \angle APD = \angle BQC$ (proven above)

$AD = BC$ (opp. sides of parallelogram $ABCD$)

$\angle ADP = \angle CBQ$ (alternate angles; $CB \parallel DA$)

$\therefore \Delta ADP \equiv \Delta CBQ$ (A.A.S.)



$\Delta ABC \parallel \Delta DEC$ (equiangular)

$$\frac{a}{a-r} = \frac{b}{h} \quad (\text{ratio of corresponding sides in similar triangles})$$

$$\therefore \frac{b}{a} = \frac{h}{a-r}$$

$$\begin{aligned}
 \text{ii) } V &= \pi r^2 h \\
 &= \pi r^2 \cdot \frac{b}{a} \cdot (a - r) \\
 &= \frac{\pi b}{a} (ar^2 - r^3)
 \end{aligned}$$

$$\text{iii) Volume of cone} = \frac{1}{3} \pi a^2 b$$

$$\frac{dV}{dr} = \frac{\pi b}{a} (2ar - 3r^2)$$

$$\frac{dV}{dr} = 0 \text{ when } 2ar = 3r^2$$

$$\text{i.e. } r=0, \text{ or } r = \frac{2a}{3}$$

when $r=0$, there's no cylinder
 $\therefore$ only test when $r = \frac{2a}{3}$ using
 the first derivative as the fn is
 cts.

r	$< \frac{2a}{3}$	$= \frac{2a}{3}$	$> \frac{2a}{3}$
$\frac{dV}{dr}$	$\frac{\pi ab}{a}$	0	$-\pi ba$
slope	/	-	\

$\therefore$ rel. max. at $r = \frac{2a}{3}$, since
 there's only 1 stat. pt in the domain $0 < r < a$, the
 rel. max. is the absolute maximum.
 $\therefore$ max. Volume of cylinder is:

$$\begin{aligned}
 V &= \pi \frac{b}{a} \left(a \frac{4a^2}{9} - \frac{8a^3}{27} \right) \\
 &= b\pi \left(\frac{12a^2 - 8a^2}{27} \right) \\
 &= \frac{4a^2 b \pi}{27}
 \end{aligned}$$

which is $\frac{4}{9}$ of the cone's volume

Q.E.D.

(END OF SECTION A)

Q5 (15 marks)

$$\text{a) } P(x) = x^3 + 2x^2 - kx + 5$$

$$P(-3) = -27 + 18 + 3k + 5 = 1$$

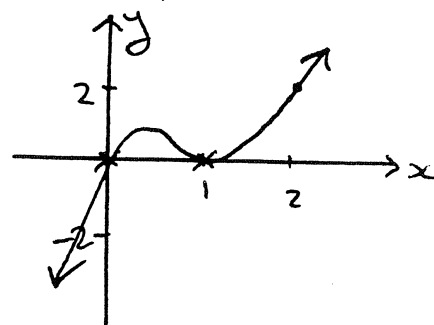
$$-4 + 3k = 1$$

$$3k = 5$$

$$\therefore k = \frac{5}{3}$$

$$\text{b) } y = x(x-1)^2$$

$$(0,0) \quad (1,0)$$



$$\text{c) } \frac{1}{\tan A + \cot B} + \frac{1}{\cot A + \tan B} = \frac{\sin(A+B)}{\cos(A-B)}$$

$$\text{L.H.S.} = \frac{1}{\tan A + \cot B} + \frac{1}{\cot A + \tan B}$$

$$= \frac{1}{\frac{\sin A}{\cos A} + \frac{\cos B}{\sin B}} + \frac{1}{\frac{\cos A}{\sin A} + \frac{\sin B}{\cos B}}$$

$$= \frac{\sin B \cos A}{\sin A \sin B + \cos A \cos B} + \frac{\sin A \cos B}{\cos A \cos B + \sin A \sin B}$$

$$= \frac{\sin A \cos B + \sin B \cos A}{\sin A \sin B + \cos A \cos B}$$

$$= \frac{\sin(A+B)}{\cos(A-B)}$$

$$= \text{R.H.S.}$$

Q.E.D.

$$\text{d) } \tan 75^\circ = \tan(30^\circ + 45^\circ) = \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3} - 1}$$

$$e) \lim_{x \rightarrow -3} \frac{2x^3 + 54}{x+3}$$

$$= \lim_{x \rightarrow -3} \frac{2(x+3)(x^2 - 3x + 9)}{x+3}$$

$$= \lim_{x \rightarrow -3} 2(x^2 - 3x + 9)$$

$$= 2 \times (9 + 9 + 9)$$

$$= 54$$

Q6. 15 marks

$$a) f(x) = x^4 + bx^2 + cx + d$$

$$f(0) = 1 \therefore \underline{d=1}$$

$$f(1) = 0$$

even $\therefore$ symmetrical about y-axis

$$0 = 2 + b + c$$

$$\therefore b + c = -2 \dots \textcircled{1}$$

$$f(-x) = x^4 + bx^2 - cx + d$$

$$f(-x) = f(x) \text{ (even)}$$

$$\therefore \underline{c=0}$$

$$\therefore \underline{b=-2} \text{ (from } \textcircled{1})$$

$$b) i) 9! = 362880$$

$$ii) 1 \times 3! \times 7! = 30240$$

$$iii) 7 \times 8! = 282240$$

$$c) 2\sin\theta - \cos\theta = 1$$

$$2\left(\frac{2t}{1+t^2}\right) - \left(\frac{1-t^2}{1+t^2}\right) = 1$$

$$4t - 1 + t^2 = t^2 + 1$$

$$4t = 2$$

$$\therefore t = \frac{1}{2}$$

$$\text{now } \tan \frac{\theta}{2} = \frac{1}{2}$$

$$\frac{\theta}{2} = 0.4636^\circ$$

$$\therefore \theta = 0.9273^\circ$$

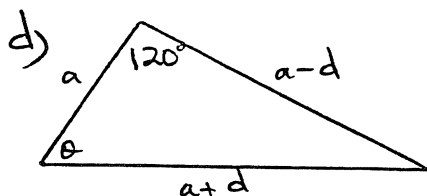
$$\text{Test } \theta = \pi,$$

$$2\sin\pi - \cos\pi = 1$$

$$0 - -1 = 1$$

$$1 = 1 \text{ true}$$

$\therefore$ solns. are $\theta = \pi$ and 0.9273°



$$(a+d)^2 = (a-d)^2 + a^2 - 2a(a-d)\cos 120^\circ$$

$$a^2 + 2ad + d^2 = a^2 - 2ad + d^2 + a^2 + (2a^2 - 2ad) \times \frac{1}{2}$$

$$0 = -4ad + a^2 + a^2 - ad$$

$$5ad = 2a^2$$

$$d \neq 0$$

$$a \neq 0$$

$$a = \frac{5d}{2}$$

$$\frac{a-d}{\sin\theta} = \frac{a+d}{\sin 120^\circ}$$

$$(\sin 120^\circ = \frac{\sqrt{3}}{2})$$

$$\sin\theta = \frac{\sqrt{3}}{2} \cdot \frac{a-d}{a+d}$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{2\frac{1}{2}d - d}{2\frac{1}{2}d + d}$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1\frac{1}{2}d}{3\frac{1}{2}d}$$

$$\sin\theta = 0.371153744$$

$$\theta = 21^\circ 47'$$

Q.E.D.

Q7 15 marks

$$a) \frac{x-2}{x-4} \geq 2$$

$$x \neq 4$$

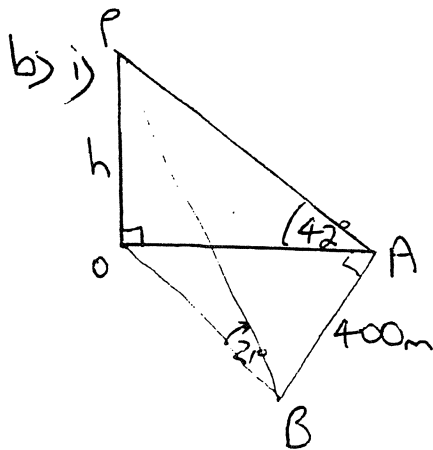
$$\frac{x-2-2(x-4)}{x-4} \geq 0$$

$$\frac{-x+6}{x-4} \geq 0$$

-	+	-	α
+	+	-	$6-x$
-	+	+	$x-4$
+		6	

$$\therefore 4 < x \leq 6$$

Q7 Continued



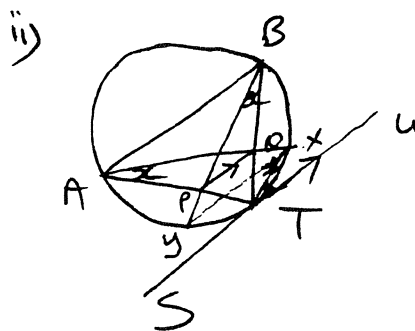
In $\triangle POB$,
 $\tan 21^\circ = \frac{h}{OB}$
 $\therefore OB = \frac{h}{\tan 21^\circ}$
 $= h \cot 21^\circ$

In $\triangle POA$,
 $\tan 42^\circ = \frac{h}{OA}$
 $OA = \frac{h}{\tan 42^\circ}$
 $OA = h \cot 42^\circ$

In $\triangle OAB$
 $OB^2 = OA^2 + AB^2$
 $h^2 \cot^2 21^\circ = h^2 \cot^2 42^\circ + 400^2$
 $h^2 (\cot^2 21^\circ - \cot^2 42^\circ) = 400^2$
 $h^2 = \frac{400^2}{(\cot^2 21^\circ - \cot^2 42^\circ)}$
 $h = \frac{400}{\sqrt{\cot^2 21^\circ - \cot^2 42^\circ}}$
QED

ii) $h = \frac{400}{\sqrt{6.786489 - 1.23346}}$
 $= \frac{400}{\sqrt{5.553}}$
 $= 169.744$
 $\approx 170\text{m (nearest metre)}$

c) i) $\angle T_u = \angle B\hat{A}T$ (angle in the alternate segment)
 $\angle T_u = \angle P\hat{Q}T$ (alternate angles; $PQ \parallel Tu$)
 $\therefore \angle B\hat{A}T = \angle P\hat{Q}T$ (both equal to $\angle T_u$)
 $\therefore ABQP$ is a cyclic quad.
 (exterior angle equals the interior opposite angle.)



Let $\angle PBQ = x$

$\therefore \angle PAQ = x$ (angles in same segment, $PABQ$ is cyclic quad.)

$\therefore \angle XT_u = x$ (angle between chord XT and ~~tangent~~ equals angle in alternate segment)

$\angle BT = \angle XT = x$ (angles in same segment.)

$\angle XT = \angle XT_u$ (both x)

$\therefore XT \parallel ST_u$ (one pair of alternate angles is equal.)